

	Steel wire for mechanical springs Part 1: Patented cold drawn unalloyed spring steel wire English version of DIN EN 10270-1	DIN EN 10270-1
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ICS 77.140.25; 77.140.65

This standard, together with DIN EN 12166, April 1998 edition, and DIN EN 10218-2, August 1996 edition, supersedes DIN 2076 and DIN 17223-1, December 1984 editions.

Stahldraht für Federn – Teil 1: Patentiert-gezogener unlegierter Federstahldraht

European Standard EN 10270-1 : 2001 has the status of a DIN Standard.

A comma is used as the decimal marker.

National foreword

This standard has been prepared by ECISS/TC 30/SC 2 'Steel wire for mechanical springs' (Secretariat: Belgium).

The responsible German body involved in its preparation was the *Normenausschuss Eisen und Stahl* (Steel and Iron Standards Committee).

DIN 50192 is the standard corresponding to EURONORM 104 referred to in clause 2 of the EN.

Amendments

DIN 2076 and DIN 17223-1, December 1984 editions, have been superseded by the specifications of DIN EN 12166, April 1998 edition, DIN EN 10218-2, August 1996 edition, and EN 10270-1.

Previous editions

DIN 2076: 1944-02, 1984-12; DIN 17223: 1955-04; DIN 17223-1: 1964-03, 1984-12.

National Annex NA

Standard referred to

(and not included in **Normative references**)

DIN 50192 Determination of depth of decarburization of steel

EN comprises 24 pages.

ICS 77.140.25; 77.140.65

English version

Steel wire for mechanical springs

Part 1: Patented cold drawn unalloyed spring steel wire

Fils an acier pour ressorts
mécaniques – Partie 1: Fils
pour ressorts en acier non
allié, patentés, tréfilés à froid

Stahldraht für Federn – Teil 1:
Patentiert-gezogener unlegierter
Federstahldraht

This European Standard was approved by CEN on 2001-02-19.

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Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Management Centre or to any CEN member.

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CEN

European Committee for Standardization
Comité Européen de Normalisation
Europäisches Komitee für Normung

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Foreword

This European Standard has been prepared by Technical Committee ECISS/TC 30 "Steel wires ", the secretariat of which is held by BSI.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by October 2001, and conflicting national standards shall be withdrawn at the latest by October 2001.

This European Standard for steel wire for mechanical springs is composed of the following parts:

- Part 1 : *Patented cold drawn unalloyed spring steel wire*
- Part 2 : *Oil hardened and tempered spring steel wire*
- Part 3 : *Stainless spring steel wire*

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom.

1 Scope

1.1 This Part of EN 10270 applies to patented cold drawn unalloyed steel wire of circular cross-section for the manufacture of mechanical springs for static duty and dynamic duty applications.

1.2 In addition to this part of EN 10270 the general technical delivery requirements of EN 10021 are applicable.

2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

EN 10002-1, *Metallic materials — Tensile test — Part 1 : Method of test (at ambient temperature)*

EN 10016-1, *Non alloy steel rod for drawing and/or cold rolling — Part 1: General requirements*

EN 10016-2, *Non alloy steel rod for drawing and/or cold rolling — Part 2: Specific requirements for general purposes*

EN 10016-4, *Non alloy steel rod for drawing and/or cold rolling — Part 4: Specific requirements for rod for special applications*

EN 10021, *General technical delivery requirements for steel and iron products*

EN 10052, *Vocabulary of heat treatments terms for ferrous products*

EN 10204, *Metallic products — Types of inspection documents*

EN 10218-1:1994, *Steel wire and wire products — General — Part 1: Test methods*

EN 10218-2:1996, *Steel wire and wire products — General — Part 2: Dimensions and tolerances*

EN 10244-2, *Non-ferrous metallic coatings on wire — Part 2: Zinc and zinc alloy coatings for steel wire and products*

CR 10261, *Iron and steel — Review of available methods of chemical analysis*

EN ISO 377, *Steel and steel products — Location and preparation of samples and test pieces for mechanical testing*

ISO 14284, *Steel and iron — Sampling and preparation of samples for the determination of chemical composition*

EU 104¹⁾, *Determination of the decarburization depth of unalloyed and low alloyed structural steels*

1) It may be agreed at the time of ordering, until this EURONORM has been adopted as a European Standard, that either this EURONORM or a corresponding national standard should be applied

3 Terms and definitions

For the purposes of this standard the following term and definition apply:

3.1

patented cold drawn wire

wire drawn to size by cold deformation of a starting material that has been subjected to a thermal treatment of patenting (see EN 10052)

4 Classification and designation

4.1 Classification

The grade of spring wire used depends on the stress level and the nature of the duty. Where springs are subjected to static stresses or infrequent dynamic loading a wire grade for static duty (S) shall be used. In the other cases with frequent or predominantly dynamic loading and where small coiling ratios or severe bending radius is required, a wire grade for dynamic duty (D) shall be used. Depending on the stress level, spring wire is manufactured in 3 tensile strength grades: Low, medium and high.

Table 1 gives an overview of the different grades.

Table 1 — Spring wire grades

Tensile strength^a	Static	Dynamic
Low tensile strength	SL	-
Medium tensile strength	SM	DM
High tensile strength	SH	DH
^a For specific applications another tensile strength may be agreed.		

4.2 Designation

For products supplied according to this standard the designation shall state in the following order:

- the term: spring wire;
- the number of this European Standard : EN 10270-1;
- the wire grade (see Table 1);
- the required nominal diameter selected from Table 3;
- the coating indicated by its abbreviation (see 6.3).

EXAMPLE: Standard designation of a steel spring wire according to this standard of spring wire grade SM, with a nominal diameter of 2,50 mm, phosphate coated :

Spring wire EN 10270-1 - SM - 2,50 ph.

5 Information to be supplied by the purchaser

The purchaser shall clearly state in his enquiry or order the product and following information :

- a) the desired quantity;
- b) the number of this European Standard: EN 10270-1;
- c) wire grade; coating and surface finish (see 6.3);
- d) the nominal wire diameter;
- e) the form of delivery and unit mass (see 6.2);
- f) the type of inspection document;
- g) any particular agreement made.

EXAMPLE 5 t spring steel wire EN 10270-1 - SM - 2,50 ph.

on spools of about 500 kg.

inspection document EN 10204-3.1.B

6 Requirements

6.1 Material

6.1.1 General

Steel spring wire shall be made from steel corresponding to EN 10016-1 with in addition

- for SL, SM and SH EN 10016-2
- for DM and DH EN 10016-4

6.1.2 Chemical composition

The chemical composition according to the heat analysis shall comply with the limit values shown in Table 2. The permissible deviation of the product analysis from the heat analysis shall be in accordance with EN 10016-2 and EN 10016-4 respectively.

Table 2 — Chemical composition, % by mass

Grade	C ^a	Si	Mn ^b	P max.	S max.	Cu max.
SL, SM, SH	0,35-1,00	0,10-0,30	0,50-1,20	0,035	0,035	0,20
DM, DH	0,45-1,00	0,10-0,30	0,50-1,20	0,020	0,025	0,12
^a Such a wide range is stipulated to accommodate the whole range of sizes. For individual sizes the carbon range is substantially more restricted. ^b The range of manganese content in the table is wide to cope with various processing situations and the broad size range. The actual figures per size shall be more restricted.						

The addition of micro-alloying elements may be agreed between the manufacturer and the purchaser.

NOTE Some diameter ranges require particular attention for residuals. Therefore no figures are mentioned for chromium, nickel, molybdenum, tin, etc. leaving room for special arrangements between purchaser and supplier, dependent on their mutual processing conditions. This is also the case for the aluminium content.

6.2 Form of delivery

The wire shall be delivered in unit packages of a coil, individualized or on carriers, or a spool or spoolless core or as straight lengths.

Unless otherwise agreed at the time of ordering, the form of delivery will be coils; straight lengths shall be supplied in bundles.

6.3 Coating and surface finish

The spring wire may be supplied phosphate coated (ph) either dry drawn or wet drawn, copper coated (cu), zinc (Z) or zinc/aluminium (ZA) coated.

Other coatings, considered as special, can be agreed between the purchaser and the supplier (see annex A).

If no specific surface finish is specified, the type of finish shall be at the manufacturer's discretion.

In addition the wire can be ordered with an oiled surface for all surface finishes.

6.4 Mechanical properties

For the tensile strength (R_m) and reduction in area after fracture (Z) the wire grades shall satisfy the values listed in Table 3. Reduction of area shall be measured only for wire diameter 0,80 mm and above.

The range of tensile strength values within a unit package shall not exceed the values of Table 4.

Table 3 — Mechanical properties ^{a b} and quality requirements for wire grades SL, SM, DM, SH and DH

1	2	3	4	5	6	7	8	9	10	11	12	13
Wire diameter d^c	Nominal size mm	Mass kg/1000 m	Tensile strength $R_m^{c d e}$ For wire grades					Minimum reduction in area after fracture Z for wire grades SL, SM, SH, DM and DH	Minimum number of twists in the torsion test for wire grades SL, SM, SH, DM and DH	Permissible depth of surface defects for wire grades DM, DH	Permissible decarburization depth for wire grades DM, DH	Wire diameter d (nominal size) mm
			SL MPa	SM MPa	DM MPa	SH MPa	DH ^f MPa					
0,05		0,0154					2800 to 3520					0,05
0,06		0,0222					2800 to 3520					0,06
0,07	± 0,003	0,0302					2800 to 3520					0,07
0,08		0,0395					2800 to 3480					0,08
0,09		0,0499					2800 to 3430					0,09
0,10		0,0617					2800 to 3380					0,10
0,11		0,0746					2800 to 3350					0,11
0,12	± 0,004	0,0888					2800 to 3320					0,12
0,14		0,121					2800 to 3250					0,14
0,16		0,158					2800 to 3200					0,16
0,18		0,200					2800 to 3160					0,18
0,20	± 0,005	0,247					2800 to 3110					0,20
0,22		0,298					2770 to 3080					0,22
0,25		0,385					2720 to 3010					0,25
0,28		0,488					2680 to 2970					0,28
0,30		0,555					2660 to 2940					0,30
0,32	± 0,008	0,631	2370 to 2650	2370 to 2650	2370 to 2650	2660 to 2940	2660 to 2940					0,32
0,34		0,713	2350 to 2630	2350 to 2630	2350 to 2630	2640 to 2920	2640 to 2920					0,34
0,36		0,799	2330 to 2600	2330 to 2600	2330 to 2600	2610 to 2890	2610 to 2890					0,36
0,38		0,890	2310 to 2580	2310 to 2580	2310 to 2580	2590 to 2870	2590 to 2870					0,38
			2290 to 2560	2290 to 2560	2290 to 2560	2570 to 2850	2570 to 2850					0,38

^g -
coiling test as specified in 7.4.3

Table 3 — Mechanical properties^{a b} and quality requirements for wire grades SL, SM, DM, SH and DH (continued)

1	2	3	4	5	6	7	8	9	10	11	12	13			
Wire diameter d^f		Mass kg/1000 m	Tensile strength $R_m^{c d e}$ For wire grades					Minimum reduction in area after fracture Z for wire grades SL, SM, SH, DM and DH	Minimum number of twists in the torsion test for wire grades SL, SM, SH, DM and DH	Permissible depth of surface defects for wire grades DM, DH	Permissible decarburi- zation depth for wire grades DM, DH	Wire diameter d (nominal size)			
Nominal size mm	Permissible Deviations mm	SL MPa	SM MPa	DM MPa	SH MPa	DH ⁱ MPa									
0,40	± 0,008	0,985	2270 to 2550	2270 to 2550	2560 to 2830	2560 to 2830	colling test as specified in 7.4.3			- ^g		0,40			
0,43		1,14	2250 to 2520	2250 to 2520	2530 to 2800	2530 to 2800						0,43			
0,45		1,25	2240 to 2500	2240 to 2500	2510 to 2780	2510 to 2780						0,45			
0,48		1,42	2220 to 2480	2220 to 2480	2490 to 2760	2490 to 2760						0,48			
0,50		1,54	2200 to 2470	2200 to 2470	2480 to 2740	2480 to 2740						0,50			
0,53	1,73	2180 to 2450	2180 to 2450	2460 to 2720	2460 to 2720	0,53									
0,56	1,93	2170 to 2430	2170 to 2430	2440 to 2700	2440 to 2700	0,56									
0,60	2,22	2140 to 2400	2140 to 2400	2410 to 2670	2410 to 2670	0,60									
0,63	2,45	2130 to 2380	2130 to 2380	2390 to 2650	2390 to 2650	0,63									
0,65	2,60	2120 to 2370	2120 to 2370	2380 to 2640	2380 to 2640	0,65									
0,70	± 0,010	3,02	2090 to 2350	2090 to 2350	2360 to 2610	2360 to 2610					0,70				
0,75		3,47	2070 to 2320	2070 to 2320	2330 to 2580	2330 to 2580					0,75				
0,80	± 0,015	3,95	2050 to 2300	2050 to 2300	2310 to 2560	2310 to 2560	25					0,80			
0,85		4,45	2030 to 2280	2030 to 2280	2290 to 2530	2290 to 2530						0,85			
0,90		4,99	2010 to 2260	2010 to 2260	2270 to 2510	2270 to 2510						0,90			
0,95		5,59	2000 to 2240	2000 to 2240	2250 to 2490	2250 to 2490						0,95			
1,00		6,17	1720 to 1970	1980 to 2220	1980 to 2220	2230 to 2470						2230 to 2470	1 % max. of wire diameter	1,5 % max. of wire diameter	1,00
1,05	6,80	1710 to 1950	1960 to 2200	1960 to 2200	2210 to 2450	2210 to 2450						1,05			
1,10	± 0,020	7,46	1690 to 1940	1950 to 2190	1950 to 2190	2200 to 2430						2200 to 2430			1,10
1,20		8,88	1670 to 1910	1920 to 2160	1920 to 2160	2170 to 2400						2170 to 2400			1,20

Table 3 — Mechanical properties ^{a b} and quality requirements for wire grades SL, SM, DM, SH and DH (continued)

1	2	3	4	5	6	7	8	9	10	11	12	13
Wire diameter d^a	Mass kg/1000 m	Tensile strength $R_m^{c,d,e}$ For wire grades						Minimum reduction in area after fracture Z for wire grades SL, SM, SH, DM and DH	Minimum number of twists in the torsion test for wire grades SL, SM, SH, DM and DH	Permissible depth of surface defects for wire grades DM, DH	Permissible decarburi- zation depth for wire grades DM, DH	Wire diameter d (nominal size)
Nominal size mm	permissible Deviations mm		SL MPa	SM MPa	DM MPa	SH MPa	DH ^f MPa	%	d	mm	mm	mm
1,25		9,63	1660 to 1900	1910 to 2140	1910 to 2140	2150 to 2380	2150 to 2380		25			1,25
1,30		10,42	1640 to 1890	1900 to 2130	1900 to 2130	2140 to 2370	2140 to 2370					1,30
1,40		12,08	1620 to 1860	1870 to 2100	1870 to 2100	2110 to 2340	2110 to 2340					1,40
1,50	± 0,020	13,90	1600 to 1840	1850 to 2080	1850 to 2080	2090 to 2310	2090 to 2310					1,50
1,60		15,8	1590 to 1820	1830 to 2050	1830 to 2050	2060 to 2290	2060 to 2290					1,60
1,70		17,8	1570 to 1800	1810 to 2030	1810 to 2030	2040 to 2260	2040 to 2260					1,70
1,80		20,0	1550 to 1780	1790 to 2010	1790 to 2010	2020 to 2240	2020 to 2240					1,80
1,90		22,3	1540 to 1760	1770 to 1990	1770 to 1990	2000 to 2220	2000 to 2220					1,90
2,00		24,7	1520 to 1750	1760 to 1970	1760 to 1970	1980 to 2200	1980 to 2200					2,00
2,10	± 0,025	27,2	1510 to 1730	1740 to 1960	1740 to 1960	1970 to 2180	1970 to 2180		22	1 % max. of wire diameter	1.5 % max. of wire diameter	2,10
2,25		31,2	1490 to 1710	1720 to 1930	1720 to 1930	1940 to 2150	1940 to 2150	40				2,25
2,40		35,5	1470 to 1690	1700 to 1910	1700 to 1910	1920 to 2130	1920 to 2130					2,40
2,50		38,5	1460 to 1680	1690 to 1890	1690 to 1890	1900 to 2110	1900 to 2110					2,50
2,60		41,7	1450 to 1660	1670 to 1880	1670 to 1880	1890 to 2100	1890 to 2100					2,60
2,80		48,3	1420 to 1640	1650 to 1850	1650 to 1850	1860 to 2070	1860 to 2070					2,80
3,00		55,5	1410 to 1620	1630 to 1830	1630 to 1830	1840 to 2040	1840 to 2040					3,00
3,20	± 0,030	63,1	1390 to 1600	1610 to 1810	1610 to 1810	1820 to 2020	1820 to 2020					3,20
3,40		71,3	1370 to 1580	1590 to 1780	1590 to 1780	1790 to 1990	1790 to 1990					3,40
3,60		79,9	1350 to 1560	1570 to 1760	1570 to 1760	1770 to 1970	1770 to 1970		16			3,60
3,80		89,0	1340 to 1740	1550 to 1740	1550 to 1740	1750 to 1950	1750 to 1950					3,80

Table 3 — Mechanical properties^{a b} and quality requirements for wire grades SL, SM, DM, SH and DH (continued)

1	2	3	4	5	6	7	8	9	10	11	12	13
Wire diameter d^d	Nominal size mm	Mass kg/1000 m	Tensile strength $R_m^{c d e}$ For wire grades					Minimum reduction in area after fracture Z for wire grades SL, SM, SH, DM and DH	Minimum number of twists in the torsion test for wire grades SL, SM, SH, and DH	Permissible depth of surface defects for wire grades DM, DH	Permissible decarburization depth for wire grades DM, DH	Wire diameter d (nominal size)
			SL MPa	SM MPa	DM MPa	SH MPa	DH ^f MPa					
4,00	± 0,030	98,6	1320 to 1520	1530 to 1730	1530 to 1730	1740 to 1930	1740 to 1930	35	16	1 % max. of wire diameter	1.5 % max. of wire diameter	4,00
4,25		111	1310 to 1500	1510 to 1700	1510 to 1700	1710 to 1900	1710 to 1900					4,25
4,50		125	1290 to 1490	1500 to 1680	1500 to 1680	1690 to 1880	1690 to 1880					4,50
4,75	± 0,035	139	1270 to 1470	1480 to 1670	1480 to 1670	1680 to 1860	1680 to 1860					4,75
5,00		154	1260 to 1450	1460 to 1650	1460 to 1650	1660 to 1840	1660 to 1840					5,00
5,30		173	1240 to 1430	1440 to 1630	1440 to 1630	1640 to 1820	1640 to 1820					5,30
5,60		193	1230 to 1420	1430 to 1610	1430 to 1610	1620 to 1800	1620 to 1800					5,60
6,00		222	1210 to 1390	1400 to 1580	1400 to 1580	1590 to 1770	1590 to 1770					6,00
6,30	± 0,040	245	1190 to 1380	1390 to 1560	1390 to 1560	1570 to 1750	1570 to 1750					6,30
6,50		260	1180 to 1370	1380 to 1550	1380 to 1550	1560 to 1740	1560 to 1740					6,50
7,00		302	1160 to 1340	1350 to 1530	1350 to 1530	1540 to 1710	1540 to 1710					7,00
7,50		347	1140 to 1320	1330 to 1500	1330 to 1500	1510 to 1680	1510 to 1680					7,50
8,00		395	1120 to 1300	1310 to 1480	1310 to 1480	1490 to 1660	1490 to 1660					8,00
8,50	± 0,045	445	1110 to 1280	1290 to 1460	1290 to 1460	1470 to 1630	1470 to 1630					8,50
9,00		499	1090 to 1260	1270 to 1440	1270 to 1440	1450 to 1610	1450 to 1610					9,00
9,50		559	1070 to 1250	1260 to 1420	1260 to 1420	1430 to 1590	1430 to 1590					9,50
10,00	± 0,050	617	1060 to 1230	1240 to 1400	1240 to 1400	1410 to 1570	1410 to 1570					10,00
10,50		680		1220 to 1380	1220 to 1380	1390 to 1550	1390 to 1550					10,50
11,00	± 0,070	746	-	1210 to 1370	1210 to 1370	1380 to 1530	1380 to 1530					11,00
12,00	± 0,080	888		1180 to 1340	1180 to 1340	1350 to 1500	1350 to 1500					12,00

Table 3 — Mechanical properties^{a,b} and quality requirements for wire grades SL, SM, DM, SH and DH (continued)

1	2	3	4	5	6	7	8	9	10	11	12	13
Wire diameter d ^f	Nominal size mm	Mass kg/1000 m	Tensile strength R _m ^{c,d,e} For wire grades					Minimum reduction in area after fracture Z for wire grades SL, SM, SH, DM and DH	Minimum number of twists in the wire grades SL, SM, SH, DM and DH	Permissible depth of surface defects for wire grades DM, DH	Permissible decarburization depth for wire grades DM, DH	Wire diameter d (nominal size) mm
			SL MPa	SM MPa	DM MPa	SH MPa	DH ⁱ MPa					
12,50		963		1170 to 1320	1170 to 1320	1330 to 1480	1330 to 1480					12,50
13,00	± 0,080	1042		1160 to 1310	1160 to 1310	1320 to 1470	1320 to 1470					13,00
14,00		1208		1130 to 1280	1130 to 1280	1290 to 1440	1290 to 1440					14,00
15,00		1387		1160 to 1260	1160 to 1260	1270 to 1410	1270 to 1410					15,00
16,00		1578		1090 to 1230	1090 to 1230	1240 to 1390	1240 to 1390					16,00
17,00	± 0,090	1782	-	1070 to 1210	1070 to 1210	1220 to 1360	1220 to 1360			1 % max. of wire diameter	1.5 % max. of wire diameter	17,00
18,00		1998		1050 to 1190	1050 to 1190	1200 to 1340	1200 to 1340					18,00
19,00		2225		1030 to 1170	1030 to 1170	1180 to 1320	1180 to 1320					19,00
20,00	± 0,100	2466		1020 to 1150	1020 to 1150	1160 to 1300	1160 to 1300					20,00

^a For intermediate values of wire diameter, the values specified for the next higher diameter shall apply (for tensile strength see c).

^b Wire with a size above 20 mm diameter is used. Where such wire is specified the parties shall agree upon the properties and requirements at the time of enquiry and order.

^c For sizes not mentioned the required strength shall be derived from the mathematical formulae given in A.4.

^d For straightened and cut lengths the tensile strength values may be up to 10 % lower, torsion values are also lowered by the straightening and cutting operation.

^e 1 MPa = 1 N/mm².

^f For diameters 0,05 mm to 0,18 mm, a restricted tensile strength range of 300 MPa within the specified range may be agreed.

^g Because of the small wire diameter, measurement of the depth of defects or depth of decarburization can only be carried out with difficulty. For this reason, no maximum value is specified for this diameter range.

^h Guideline values; not mandatory for acceptance.

Table 4 — Permissible tensile strength range within a single unit package in MPa

Diameter range (mm)	SL, SM, SH	DM, DH
$d < 0,80$	150	150
$0,80 \leq d < 1,60$	120	100
$1,60 \leq d$	120	70

The requirements apply to unit packages the mass of which in kilograms does not exceed the value of $250 \times d$ (d = wire diameter in mm) or a maximum value of 1 000 kg.

In cases of coils of greater mass, appropriate agreements shall be made.

6.5 Technological properties

6.5.1 Coiling test

In order to assess the uniformity of the wire in the coiling deformation and its surface condition, the coiling test shall be carried out on wire with a diameter up to 0,70 mm for the grades DM, SH and DH.

In the test as described further in 7.4.3 the test piece shall exhibit a defect-free surface without splits or fracture, a uniform pitch of the turns after coiling and a fair dimensional regularity of its diameter.

NOTE Although the usefulness of the coiling test is not generally recognized, it has been retained since it offers the possibility of revealing internal stresses. If doubtful test results are obtained the wire concerned should not be rejected immediately but efforts should be made by the parties concerned to elucidate the cause.

6.5.2 Torsion test

For assessing the deformability, fracture behaviour and surface condition the torsion test shall be carried out for all wire grades on wires in the nominal diameter range over 0,70 mm to 10 mm. The minimum number of turns specified in Table 3 for diameters up to 7,00 mm is mandatory. For wires exceeding this size they shall be taken only as indicative values.

In executing the test according to 7.4.5 the required number of turns shall be achieved before the test piece fractures. The fracture of the torsion test piece shall be perpendicular to the wire axis (see EN 10218-1:1994 - Types 1a, 2a or 3a).

Spring back resilience cracks or spring back fractures ("spoon" or "secondary" fractures) are not considered in the evaluation. In each case a uniform distortion of both fragments shall be present although the pitch of the turns need not be the same in the two parts. In the case of grade DH no surface cracks visible to the naked eye shall be present after the torsion test (only type "1a" fracture is acceptable).

6.5.3 Wrapping test

The wrapping test may be applied (see 7.4.4) to wire with a diameter equal or less than 3,00 mm. The wire shall not show sign of fracture when close wrapped eight turns around a mandrel of a diameter equal to that of the wire.

6.5.4 Bend test

Where requested the bend test may be applied to wire with a diameter greater than 3,00 mm. The wire shall withstand the test without any sign of failure.

NOTE In some applications the material is severely deformed by bending. Such is the case for extension springs with tight hooks, springs with bend on legs, spring wire forms, etc. In such cases the bend test provides for a wire test very close to the actual use.

6.6 Supply conditions of wire on coils/reels and spools

6.6.1 General

The wire of a unit package shall consist of one single length of wire originating from only one heat. For grades DM and DH only the welds prior to the last patenting operation are allowed; all other welds shall be removed, or - if so agreed upon - properly marked.

For the wire grades SL/SM/SH welds are allowed at the size of the last patenting treatment. For other welds, the treatment of the welds shall be subject to an arrangement between the parties involved, depending on wire diameter and application.

6.6.2 Coil size

The coil internal diameter of coils shall at least satisfy the values given in Table 5, unless otherwise agreed.

Table 5 — Wire diameter and associated minimum coil internal diameter

Wire diameter ^a (mm)	Minimum internal diameter (mm)
$0,25 \leq d < 0,28$	100
$0,28 \leq d < 0,50$	150
$0,50 \leq d < 0,70$	180
$0,70 \leq d < 1,60$	250
$1,60 \leq d < 4,50$	400
$4,50 \leq d$	500
^a For wire diameter below 0,25 mm specific agreements shall be made between the parties.	

6.6.3 Cast of wire

The wire shall be uniformly cast. Unless otherwise specified the wap diameter of wire supplied in coils/reels may expand when the binding wires are removed, but should usually not retract to an internal diameter smaller than the original cast diameter other than by agreement between supplier and purchaser. The expansion shall be approximately even within a single package and within all the units in a production batch.

6.6.4 Helix cast of wire

The wire shall be dead cast. The requirement shall be considered fulfilled in the case of wire finer than 5,00 mm in diameter if the following condition is satisfied.

An individual wap taken from a coil/reel or bobbin and freely hung on hook may show an axial displacement " f " at the ends of the wap (see Figure 1); this displacement " f " shall not exceed a value given by the following equation :

$$f \leq \frac{0,2D}{\sqrt[4]{d}}$$

where

f is the axial displacement in mm

D is the diameter of a free wap in mm

d is the diameter of the wire in mm

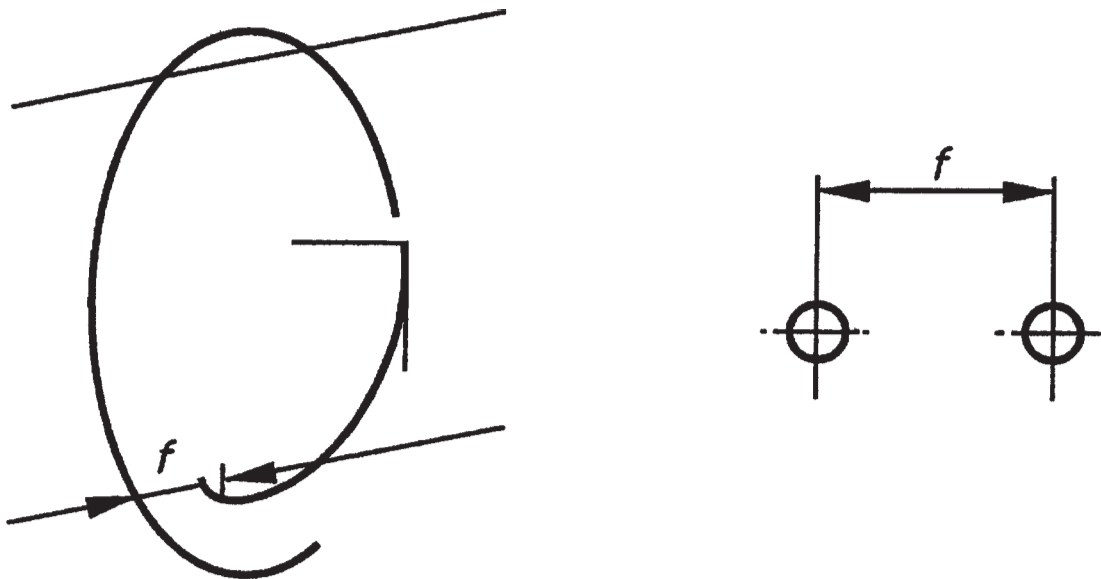


Figure 1 — Helix cast of wire

6.6.5 Other tests for cast of wire

Where appropriate other methods for testing the cast, as specified in EN 10218-1, may be agreed at the time of enquiry and order.

6.7 Surface quality

6.7.1 The surface of the wire shall be smooth and as free as possible from grooves, tears, rust and other surface defects, which have a noticeable adverse effect on the application of the wire.

6.7.2 Surface quality tests (see 7.4.7 and 7.4.8) shall be applied to wires intended for use in dynamic duty springs only (DM and DH).

- The radial depth of the seams or other surface defects shall be not greater than 1% of the nominal diameter of the wire.
- The cross-section of spring wire grades DM and DH shall show no completely decarburized layer. Partial decarburization as indicated by grain boundary ferrite of an amount in excess of that present in the main portion or "core" of the section, shall not have a radial depth greater than 1,5% of the nominal diameter of the wire.

6.7.3 In the case of zinc or zinc/aluminium coated spring wire the amount of zinc or zinc/aluminium on the wire surface shall satisfy the minimum values specified in Table 6.

Other levels may be agreed between supplier and purchaser. The adherence of the coating shall be tested by a wrapping test (see 7.4.11) according to EN 10244-2.

NOTE The usual coating processes may alter the properties of the steel wire. The ductility and endurance of the wire may thereby be reduced so that one cannot guarantee for zinc coated spring steel wire the same torsion values or expect the same dynamic performance (DM and DH) as for the respective uncoated material.

Table 6 — Minimum required zinc or zinc/aluminium coating mass

Diameter (mm)	Coating mass g/m ² ^a
0,20 ≤ d < 0,25	20
0,25 ≤ d < 0,40	25
0,40 ≤ d < 0,50	30
0,50 ≤ d < 0,60	35
0,60 ≤ d < 0,70	40
0,70 ≤ d < 0,80	45
0,80 ≤ d < 0,90	50
0,90 ≤ d < 1,00	55
1,00 ≤ d < 1,20	60
1,20 ≤ d < 1,40	65
1,40 ≤ d < 1,65	70
1,65 ≤ d < 1,85	75
1,85 ≤ d < 2,15	80
2,15 ≤ d < 2,50	85
2,50 ≤ d < 2,80	95
2,80 ≤ d < 3,20	100
3,20 ≤ d < 3,80	105
3,80 ≤ d ≤ 10,00	110
^a The requirements for the zinc coating correspond to class C of EN 10244-2	

6.8 Dimensions and dimensional tolerances

6.8.1 Dimensional tolerances

— Coiled wire

The tolerances on the diameter are specified in Table 3.

This is based on EN 10218-2:1996

- T5 for diameters below 0,80 mm;
- T4 for diameters from 0,80 mm up to 10,00 mm;
- T3 for diameters over 10,00 mm.

Where the required tolerance level is different from that mentioned in Table 3, it shall be specified at the time of order.

— Wire in straightened and cut lengths.

The requirements for length tolerances and straightness are as in EN 10218-2:1996. The tolerance on the nominal length shall only be in plus keeping the same tolerance range (Table 7).

Table 7 — Tolerances on the length of cut lengths

Nominal length L (mm)	Tolerance		
	Class 1	Class 2	Class 3
$L \leq 300$	+ 1,0 mm 0 mm		
$300 < L \leq 1000$	+ 2,0 mm 0 mm	+ 1 % 0 %	+ 2 % 0 %
$1000 < L -$	+ 0,2 % 0 %		

The tolerance for the diameter of the wire after straightening needs to be wider to cope with the increase of the section as a result of some straightening process. The acceptable level is shown in Table 8.

6.8.2 Out of roundness

The difference between the maximum and minimum diameter of the wire at the same cross section, shall not be more than 50 % of the total permissible deviation specified in Table 3.

Table 8 — Diameter tolerance for straightened and cut lengths

Nominal diameter in mm	Tolerance in mm	
	Lower tolerance	Upper tolerance
$0,05 \leq d < 0,12$	- 0,005	+ 0,007
$0,12 \leq d < 0,22$	- 0,005	+ 0,008
$0,22 \leq d < 0,26$	- 0,005	+ 0,009
$0,26 \leq d < 0,37$	- 0,006	+ 0,012
$0,37 \leq d < 0,47$	- 0,008	+ 0,015
$0,47 \leq d < 0,65$	- 0,008	+ 0,018
$0,65 \leq d < 0,80$	- 0,010	+ 0,022
$0,80 \leq d < 1,01$	- 0,015	+ 0,030
$1,01 \leq d < 1,35$	- 0,020	+ 0,040
$1,35 \leq d < 1,78$	- 0,020	+ 0,045
$1,78 \leq d < 2,01$	- 0,025	+ 0,055
$2,01 \leq d < 2,35$	- 0,025	+ 0,060
$2,35 \leq d < 2,78$	- 0,025	+ 0,065
$2,78 \leq d < 3,01$	- 0,030	+ 0,075
$3,01 \leq d < 3,35$	- 0,030	+ 0,080
$3,35 \leq d < 4,01$	- 0,030	+ 0,090
$4,01 \leq d < 4,35$	- 0,035	+ 0,100
$4,35 \leq d < 5,01$	- 0,035	+ 0,110
$5,01 \leq d < 5,45$	- 0,035	+ 0,120
$5,45 \leq d < 6,01$	- 0,040	+ 0,130
$6,01 \leq d < 7,12$	- 0,040	+ 0,150
$7,12 \leq d < 7,67$	- 0,045	+ 0,160
$7,67 \leq d < 9,01$	- 0,045	+ 0,180
$9,01 \leq d < 10,01$	- 0,050	+ 0,200
$10,01 \leq d < 11,12$	- 0,070	+ 0,240
$11,12 \leq d < 12,01$	- 0,080	+ 0,260
$12,01 \leq d < 14,52$	- 0,080	+ 0,300
$14,52 \leq d < 17,34$	- 0,090	+ 0,350
$17,34 \leq d < 18,37$	- 0,090	+ 0,370
$18,37 \leq d < 20,01$	- 0,100	+ 0,400

7 Testing and inspection

7.1 Inspection and inspection documents

Products conforming to this standard shall be delivered with specific testing (see EN 10021) and the relevant inspection document (see EN 10204) agreed at the time of enquiry and order.

The inspection document shall include the following information :

- heat analysis;
- result of the tensile test (R_m and Z);
- result of the torsion test (N_t);
- actual wire diameter;
- coating mass (where applicable);
- results of optional tests agreed.

7.2 Extent of testing for specific inspection

The extent of testing shall be in accordance with Table 10.

7.3 Sampling

Sampling and testing preparation shall be in accordance with EN ISO 377 and ISO 14284. Samples shall be taken at the end of the units. Table 10/column 8 gives further details.

7.4 Test methods

7.4.1 Chemical composition

Unless otherwise agreed at the time of ordering the choice of a suitable physical or chemical method of analysis for the determination of product analysis shall be at the discretion of the supplier.

In cases of dispute the analysis shall be carried out by a laboratory approved by the two parties. The method of analysis to be applied shall be agreed upon, where possible in accordance with CR 10261.

7.4.2 Tensile test

The tensile test shall be carried out according to EN 10002-1, on samples with the full cross-section of the wire. For the calculation of the tensile strength the actual cross-section based on the actual wire diameter is applied.

7.4.3 Coiling test

The coiling test shall be carried out in the following manner: A test piece - approximately 500 mm in length - shall be closely wound, under slight but reasonably uniform tension on a mandrel three to three and a half times the nominal diameter. The mandrel diameter shall however be at least 1,00 mm. The close coil shall be stretched so that after releasing the stress it sets to approximately three times its original length.

The surface condition of the wire and the regularity of the spring pitch and individual windings shall be inspected with the test piece in this condition.

7.4.4 Wrapping test

The wrapping test shall be carried out according to EN 10218-1; the wire shall be wrapped 8 turns around a mandrel with a diameter equal to the wire diameter.

7.4.5 Torsion test

For the torsion test, the test piece shall be clamped into the device in such manner that its longitudinal axis is aligned to the axis of the clamping heads and the test piece remains straight during the test. One clamping head shall be turned at as uniform a speed of rotation as possible (not exceeding one rotation per second) until the test piece fractures. The number of complete rotations of the turning clamping head is counted. The free length between grips shall be uniformly $100 \times d$ (d = nominal diameter of the wire) with a maximum of 300 mm.

7.4.6 Bending test

For the bending test, a wire sample of sufficient length shall be bent in U form around a mandrel with a diameter equal to twice the wire diameter for sizes above 3,00 mm to 6,50 mm and equal to three times the wire diameter for sizes above 6,50 mm. For practical reasons the wire shall be deemed to have met the requirements of this standard if it withstands bending around a mandrel smaller than the one specified. In carrying out the test the wire shall be free to move longitudinally in the forming device.

7.4.7 Surface defects

Testing for surface defects shall be carried out on test pieces from the ends of the wire units after deep etching or on microsections. It may be agreed for wire diameters below 2,00 mm at the time of order, that microscopic testing be carried out immediately after the last heat treatment.

The deep etch test shall be carried out according to EN 10218-1.

In cases where the sensitivity of eddy current testing is adequate, this method may be used by agreement.

In cases of dispute the result of measurement on the microsection applies.

7.4.8 Decarburization

Decarburization shall be inspected by microscope in accordance with EU 104 on a transverse metallographic test piece, suitably etched with a magnification of $\times 200$. The depth of decarburization is considered as being the mean of 8 measurements at the ends of four diameters located at 45° to each other, starting from the zone of maximum decarburization and avoid starting from a defective zone. In the calculation of the above mean value, any measuring point of the seven remaining situated in a local surface defect, shall not be taken into account in the calculation.

It may be agreed for wire diameters less than 2,00 mm at the time of ordering that testing be carried out immediately after the last heat treatment.

7.4.9 Diameter

The diameter shall be measured using limit gauges, a micrometer or any other appropriate method. The out of roundness shall be determined as the difference between the maximum and minimum diameters at any one cross-section. Below 0,65 mm the relative value of individual measurement (see A.3) shall be taken into consideration because the measurements are situated at the limit of the technical capability for the instruments.

7.4.10 Zinc and zinc/aluminium coating

The zinc or zinc/aluminium coating shall be measured according to EN 10244-2 by the volumetric method or the gravimetric method.

7.4.11 Adherence of coating

Adherence of zinc or zinc/aluminium coating shall be tested for wire diameters up to 5 mm in accordance with EN 10244-2 by a wrapping test on a mandrel of $3 \times d$.

7.5 Retests

Retests shall be performed according to EN 10021.

8 Marking and packaging

Each unit shall be properly marked and identified so as to permit traceability and reference to the inspection documents.

The labels shall withstand normal handling and contact with oil; they shall show the information according to Table 9. Other information shall be the subject of an arrangement between the parties.

Wire shipments shall be suitably protected against mechanical damage and/or contamination during transport.

Table 9 — Information on the labels ^a

Designation	+
Manufacturer	+
Nominal diameter	+
Spring wire grade	+
Surface finish	(+)
Cast number	(+)
Identification number	+
Coating	(+)

^a The symbols in the table mean :

+ The information shall be mentioned on the labels

(+) The information shall be mentioned on the labels if so agreed

Table 10 — Extent of testing and sampling for specific inspection and summary of the information on test procedure and requirements

	1	2	3	4	5	6	7	8	9	10	
	Test method	Applies to wire grades	^a	Test unit	Number of products per test unit	Number of samples per product	Number of test pieces per sample	Sampling	Test procedure acc. to	Requirements See	
1	Product analysis	all	0 ^b	Quantity supplied per heat	1	1	1	ISO 14284	7.4.1	6.1.2 ^b	
2	Tensile test R_m Z	All $\geq 0,80$ mm	m	Quantity supplied per production batch ^d	10 % ^c	1	1	Test pieces taken from the ends of the coils	7.4.2	6.4 ^d	
3	Coiling test	DM, SH, DH $\leq 0,70$ mm	o		The scope of testing shall be agreed on ordering				7.4.3	6.5.1	
3a	Wrapping test	All ≤ 3 mm	o		10 % ^c		1		7.4.4	6.5.3	
3b	Bending test	All ≤ 3 mm	o						7.4.6	6.5.4	
4	Torsion test ^e	All	m						7.4.5	6.5.2 ^d	
5	Testing for Freedom from twists	All	m	10 % ^c	1	1		6.6.3 6.6.4	6.6.3 6.6.4		
6	Testing for Surface defects	DM, DH	m					7.4.7	6.7 ^d		
7	Testing for decarburization	DM, DH	m					7.4.8	6.7.2 ^d		
8	Check on dimensions	All	m	100 %	To be agreed at the order			7.4.9	6.8 ^d		
9	Testing of coating mass	Z and ZA	o	To be agreed at the order			7.4.10	6.7.3			
9a	Coating adherence	Z and ZA $d \leq 5$ mm	m	10 % ^c	1	1	7.4.11	6.7.3			

^a m (= mandatory): the test is to be carried out in each case / o (= optional): the test is carried out only if so agreed at the order.

^b The results of the cast analysis for the elements listed in Table 1 for the grade concerned shall be notified to the customer in all cases.

^c 10 % of the wire units in the production batch, at least 2 but no more than 10 coils/reels or spools.

^d A production batch is defined as a quantity of production originating from the same cast, which has been subjected to the same conditions of heat treatment, and drawn with the same reduction in cross-section and with the same surface finish.

Only for diameters over 0.70 and up to 10 mm

^a m (= mandatory): the test is to be carried out in each case / o (= optional): the test is carried out only if so agreed at the order.

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^d A production batch is defined as a quantity of production originating from the same cast, which has been subjected to the same conditions of heat treatment, and drawn with the same reduction in cross-section and with the same surface finish.

^e Only for diameters over 0,70 and up to 10 mm.

Annex A (informative)

Additional information

A.1 Definition of surface condition of the wire

A.1.1 Drawing condition

All cold drawn spring wire is commonly shaped by drawing. A distinction as to the method of drawing can be made between:

- dry drawn (d) : drawn through pulverized lubricants such as soap, stearates or similar substances;
- paste drawn (ps) : drawn through very viscous greases of mineral oil base, tallow, synthetic waxes or similar substances;
- grey bright (gr) : drawn through rape oil, thin-bodied mineral oils or similar substances;
- wet drawn (w) : drawn through aqueous emulsions of greases or oil emulsions;
- liquor finished (l) : drawn through aqueous solutions with or without addition of metal salts.

A.1.2 Surface treatment

The surface for spring wire generally has a coating for facilitating wire drawing and spring forming. Exceptionally the material will be uncoated. Common surface coatings are :

- bright (b) : without any special coating : standard borax coating or lime-coating may be applied;
- phosphate coated (ph) : the wire has been treated in a solution to form a metal-phosphate layer on the surface;
- reddish (rd) : the surface is covered with a thin copper, generally a conversion coating;
- copper plated (cu) : the surface is covered with a (uniform) thick copper coating;
- galvanised (Z) : the surface is covered with a zinc coating;
- zinc/aluminium coated (ZA) : the surface is covered with a Zn 95/Al 5 coating;
- yellow coated (y) : this applies only to liquor finished products whereby a mixture of tin salts and copper salts are added at the liquor finishing;
- white (liquor finished) (wh) : this applies to liquor finished products whereby tin salts are added for liquor finishing.

A.1.3 Abbreviations

- When no particular drawing condition is required abbreviations for the surface coating only shall be used (see A.1.2) by adding it to the wire size.

EXAMPLE : 2,5 mm phosphate coated spring wire : 2,5 ph

Depending on the size the wire will be in the dry drawn (d) or wet drawn (w) condition

- For other drawing conditions or when the purchaser explicitly wishes to obtain a wet drawn or a dry drawn surface condition it shall be indicated by a combination of the abbreviation of the coating followed by the abbreviation for the drawing condition.

EXAMPLE : 3,0 mm grey bright phosphate coated spring wire : 3,0 ph gr
1,5 mm wet drawn reddish spring wire : 1,5 rd w.

A.2 Physical characteristics at room temperature

A.2.1 Modulus of elasticity and shear modulus

The modulus of elasticity is assumed to be 206 GPa and the shear modulus 81,5 GPa .

A.2.2 Density

Unless specifically measured the density of the steel wire is assumed to be 7,85 kg/dm³.

A.3 Accuracy of measuring instruments

In order to guarantee the accuracy of the values measured the accuracy of the measuring instrument, should be 10 times higher than the allowable tolerance for the measured values.

For diameters below 0,65 mm such instruments are not industrially available. Nevertheless because of the impact of the real diameter on the spring characteristics, tolerances of 3,5 and 8µm are specified. This means that everything possible should be done to keep all parameters, which can influence the accuracy, constant - such as temperature, dust, etc. Also each value can only be attributed a relative value. However, practice shows that by taking several measurements one gets a fair indication of the exact value.

A.4 Tensile strength formula

Where smaller wire diameters than those mentioned in Table 3 appear to be necessary for static grades the tensile strength shall be calculated from the following formulae :

- For grade SL : $R_{av} = 1845 - 700 \log d$

- For grade SM : $R_{av} = 2105 - 780 \log d$

where

d : diameters in mm

R_{av} : the average tensile strength in MPa .

The range shall be the same as specified in Table 3 for wire grade DH for the same wire diameter.

DM grade shall have the tensile strength as SM; for SH the values of DH apply.

A.5 Indication for the use of cold drawn spring steel wire

Table A.1 gives indicative information for the use of the various spring steel wire grades:

Table A.1

Spring wire grades	To be used for
SL	Tension, compression or torsion springs, which are predominantly subjected to low static stress.
SM	Tension, compression or torsion springs, which are subjected to medium high static stresses or rarely to dynamic stresses.
DM	Tension compression or torsion springs, which are subjected to medium high dynamic stresses. Also for wire forms which require severe bending.
SH	Tension, compression or torsion springs, which are subjected to high static stresses or slightly dynamic stresses.
DH	Tension, compression or torsion springs or wire forms, which are subjected to high static stresses or medium level dynamic stresses